

SIAI

Practical class #1

Services, Users and TCP/IP Networking

About me?

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- Office
- Attendance schedule:
Wednesday, 12:00-14:00
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Please make an appointment/warn before showing up, to avoid queues and frustration.

Operating system

- Linux:
 - Debian
- Notes:
 - Avoid graphical interface
 - Avoid graphical or ncurses interfaces for configurations
 - Use configurations files and command line tools
 - Work in groups of two students
- We will use your PC as a hypervisor to run a virtual machine using VMware Workstation

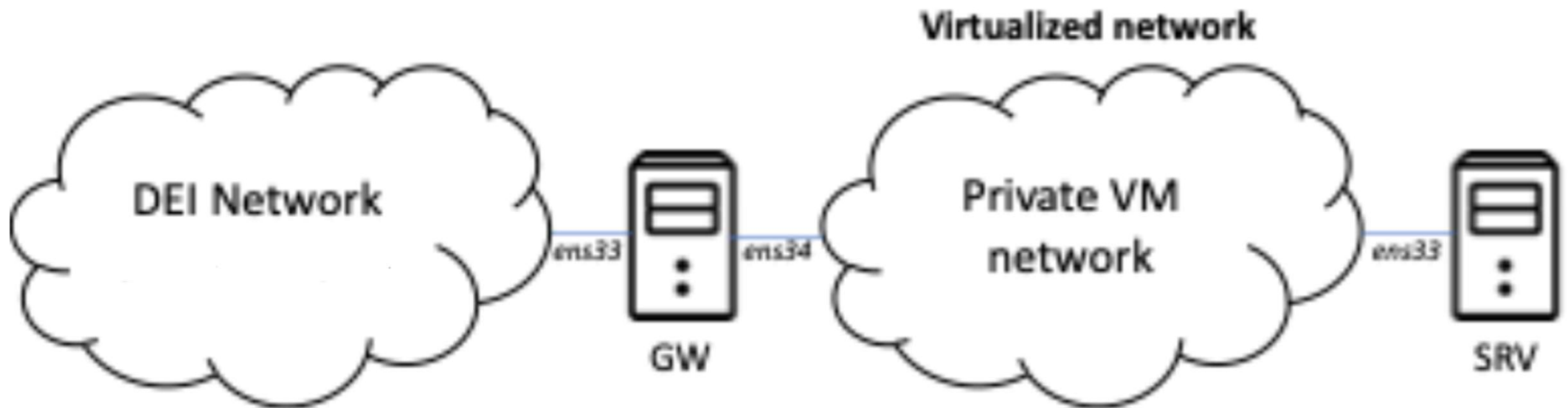
Register your groups!!!

<https://shorturl.at/D1uk7>

<https://we.tl/t-zRV3kGdp3SahmE0z>

Our setup

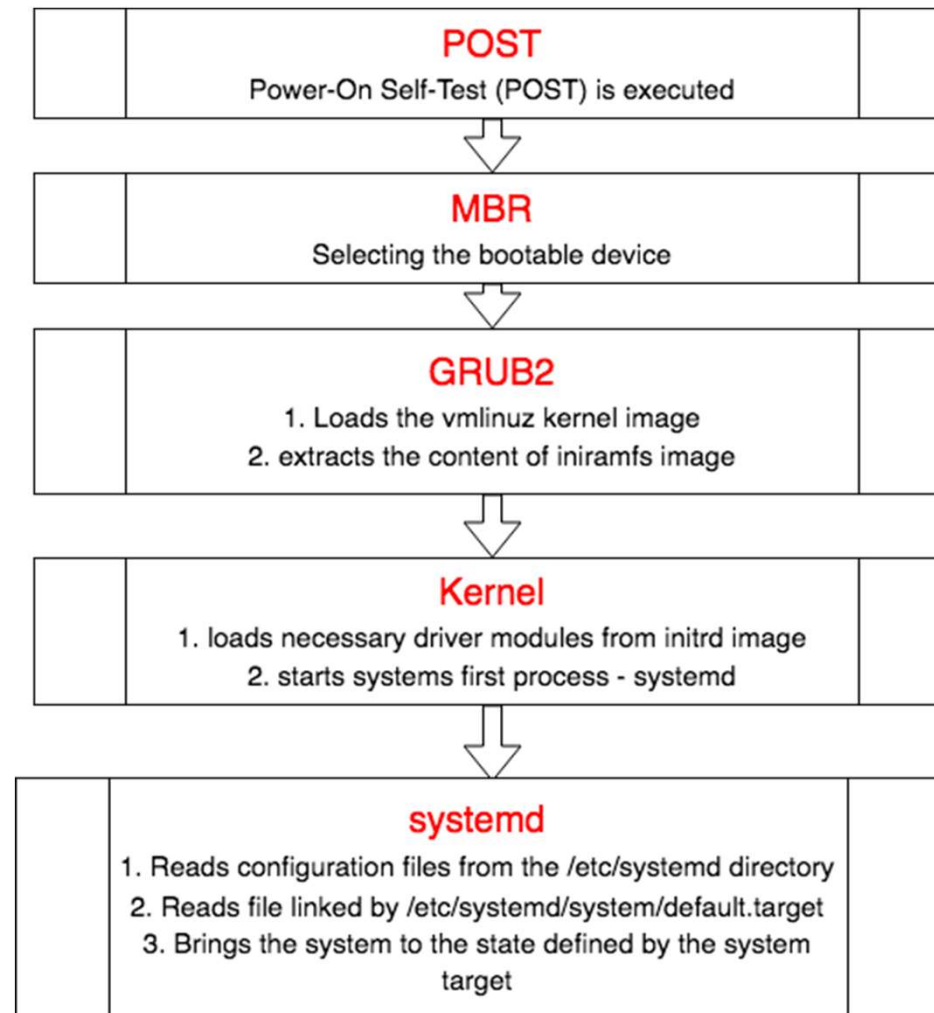
This is what you get when you log into the hypervisor



- The interfaces on the private VM network have no configured IPs
- Interface *ens33* on the gateway (GW) is configured by DHCP- NAT or Bridged
- The credentials for each VM are *root/password*.

Managing services

- **systemd**: system and service manager in most Linux distributions, is backward compatible with SysV init scripts



Managing services (2)

- Advanced Package Tool (APT), the main command-line package manager for Debian and its derivatives.

Example commands:

```
# apt update  
# apt upgrade  
# apt install httpd
```

- ***Systemctl***: tool to control systemd services, a replacement of old SysV init system management

Example commands:

```
# systemctl enable sendmail  
# systemctl start dovecot  
# systemctl rescue
```

Users and Groups

- **groupadd**: create a new group
- **useradd**: create a new user or update default new user information
- **chfn**: change your finger information
- **passwd**: update user's authentication tokens
- **id**: print real and effective user and group IDs

Example commands:

```
# groupadd gsr  
# adduser -u 1000 -g 2000 my_user  
# passwd my_user  
# id my_user
```

TCP/IP Networking

Configurations are stored in */etc/network/interfaces*

An example:

```
auto eth0
iface eth0 inet static
    address 192.0.2.7/24
    gateway 192.0.2.254
```

TCP/IP Networking (2)

- **ifconfig**: configure a network interface (old school, often deprecated)
- **ip**: show/manipulate routing, devices, policy routing and tunnels
- **route**: show/manipulate the IP routing table
- **netstat**: print network connections, routing tables, interface statistics, masquerade connections, multicast memberships
- **ping, ping6** - send ICMP ECHO_REQUEST to network hosts
- **traceroute**: print the route packets trace to network host

NFTables

nftables is a packet filtering framework in Linux that replaces **iptables**, **ip6tables**, **arptables**, and **ebtables**. It provides a more unified, efficient, and flexible approach to network filtering, NAT (Network Address Translation), and packet classification.

```
nft add table nat
nft add chain nat postrouting { type nat hook postrouting
priority 100 \; }
nft add rule nat postrouting oif ens33 masquerade

nft list ruleset
nft list ruleset > /etc/nftables.conf
nft -f /etc/nftables.conf
```

**However, is not enough. You need to activate packet forwarding
(can you find how?)**